

Declared controls and fresh-influent components

Raw connected response

χ_{raw}

$m, c_1, \dots, c_5$

g_E, g_U, M_{cl}

Physical relations
Mixer and inventories
Clarifier balances
Separation bounds
Inventory bounds
non-negativity

Empirical closure
Logarithmic model
Positive overflow solids
 $t_X^\top g_E = q_E \hat{X}_{E,\log}$

Joint nearest-response projection
Minimum standardized projection
Physical relations and fitted closure imposed together

Independent projection audit
Feasibility, non-negativity, and optimality residuals

Pass

Audited projected response $\hat{\chi}$
Used for the objective and trust screening

Fail

Projection fails
or audit fails
Retain status